

The role of maternal age on stillbirth: a multi-country analysis of more than 180 million births

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Introduction

Stillbirth remains a major global public health challenge, with burden concentrated in low- and middle-income settings (LMICs).^{1,2} While maternal age is a recognised risk factor, evidence across the full age spectrum and diverse contexts remains limited.^{1,3} Understanding how stillbirth risk varies with maternal age across settings is critical to addressing inequities in maternal and newborn outcomes.

This study used multi-country data from diverse settings to examine stillbirth by maternal age. It assessed birth distribution, stillbirth risk, and the proportion of stillbirth attributable to maternal age.

Methods

We conducted a pooled analysis of 42 datasets from diverse settings, including 180,912,564 livebirths and 1,015,385 stillbirths, stratified by region to assess variations across contexts. (Fig 1)

We estimated the distribution of births, stillbirth rate (SBR per 1,000 total births), rate ratio (RR), and the proportion of stillbirth attributed to maternal age (PAF), using ages 25–29 years as the reference group.

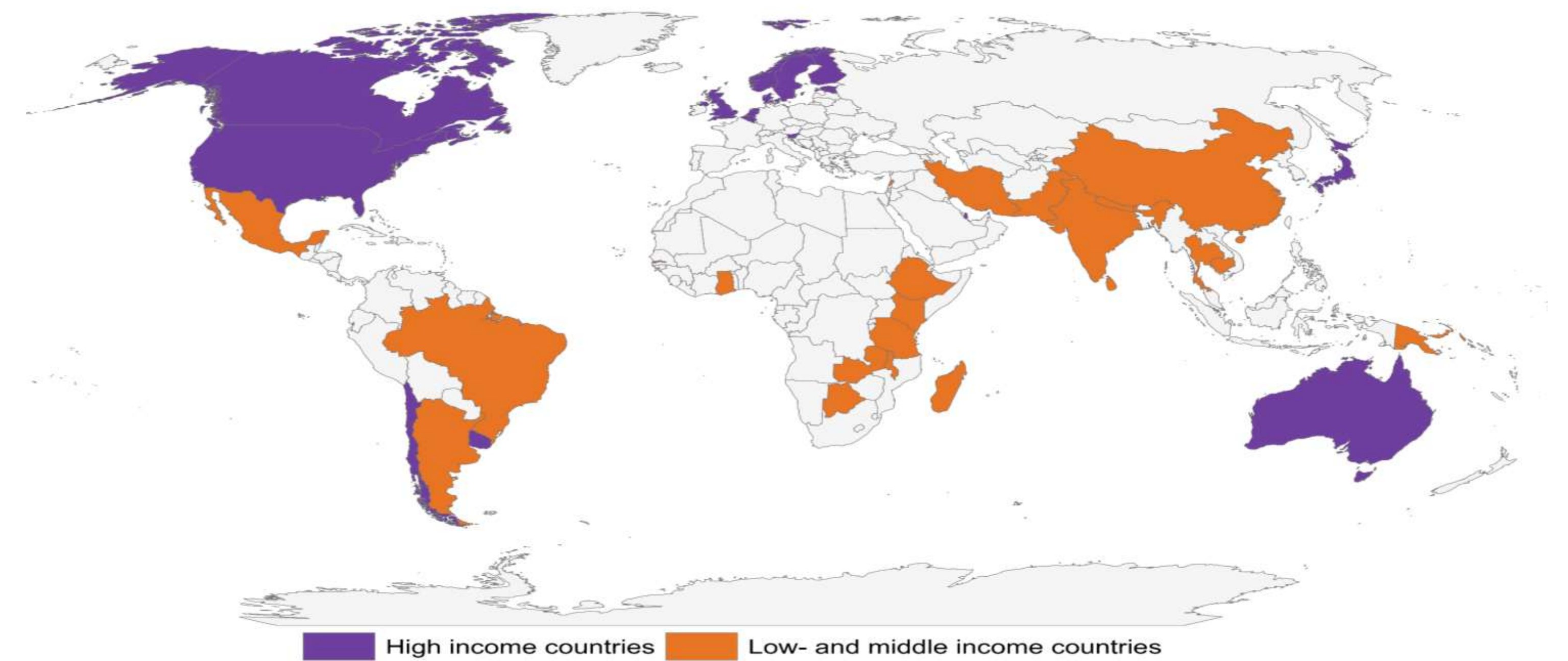


Fig 1: The geographical representation of the datasets

Results

- 70–75% of births occurred between ages 20–34 years across the regions.
- Births among adolescents (≤ 19 years) were more prevalent in LMICs (14.6%) compared to HICs (6.2%).
- Births among mothers aged ≥ 40 years accounted for 3.4% across the regions.

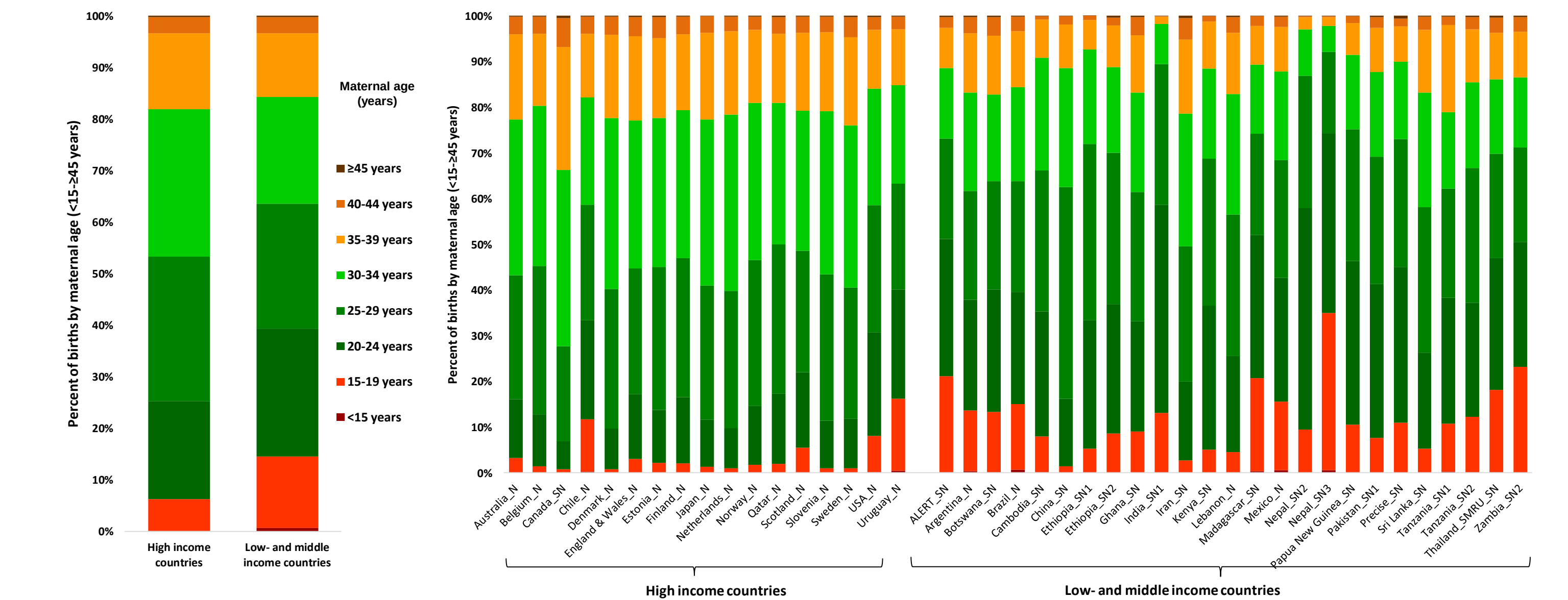
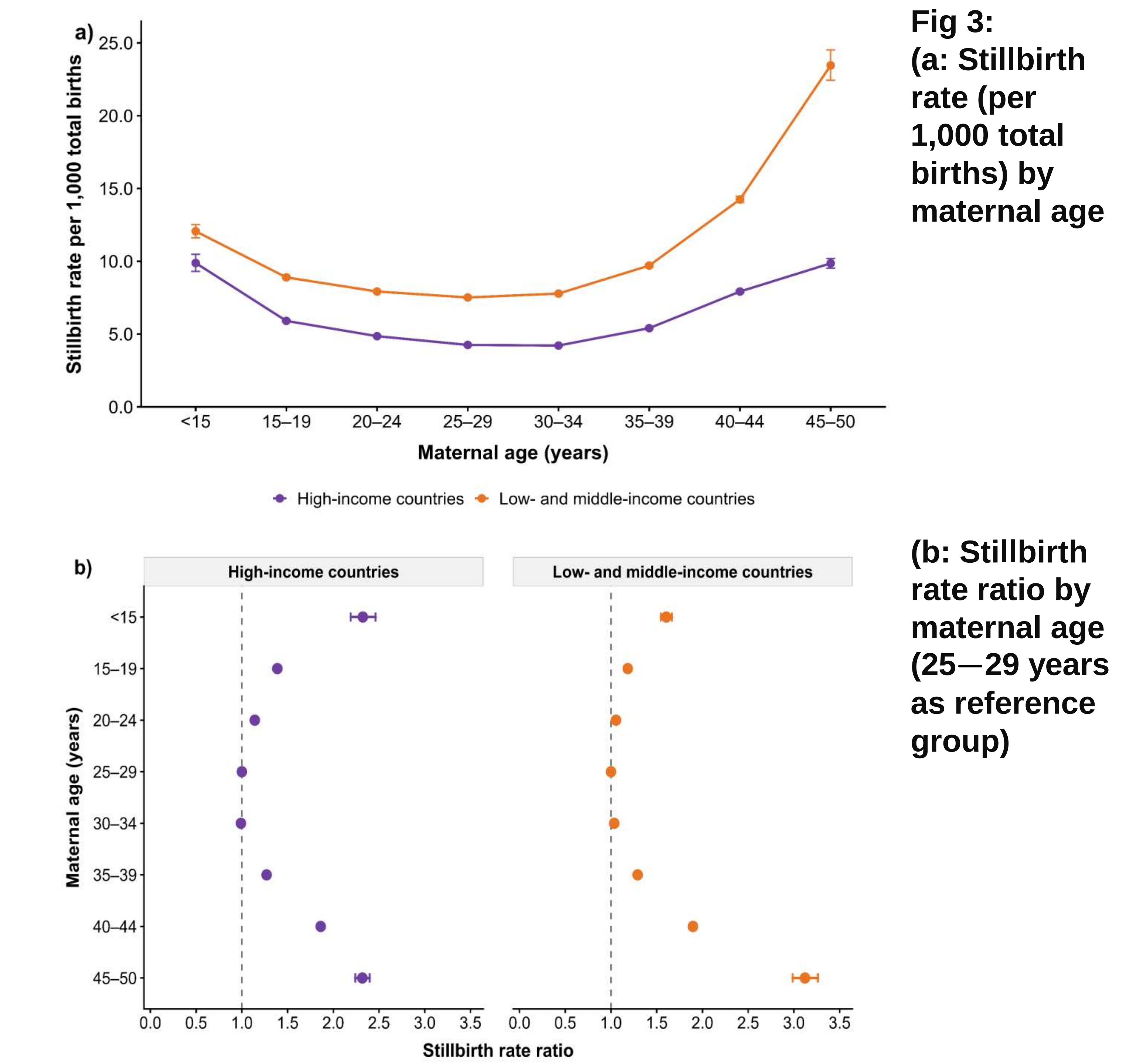


Fig 2: The distribution of births by maternal age (5-year groups) across the regions

- Stillbirth rate and ratio followed a J-shaped pattern across the regions, with the lowest rates between ages 20–34 years and the highest among adolescents (≤ 19 years) and mothers aged ≥ 40 years. (Fig 3)
- Among adolescents (≤ 19 years), SBR (per 1,000 total births) was 4.76 (95% CI: 4.75–4.77) in HICs versus 8.41 (95% CI: 8.38–8.44) in LMICs. Compared to mothers aged 25–29 years, the stillbirth rate ratio was 1.40 (95% CI: 1.39–1.41) in HICs and 1.20 (95% CI: 1.19–1.21) in LMICs.

- Among mothers aged ≥ 40 years, SBR (per 1,000 total births) was 8.06 (95% CI: 7.98–8.14) in HICs versus 14.84 (95% CI: 14.63–15.05) in LMICs. Compared to mothers aged 25–29 years, the stillbirth rate ratio was 1.89 (95% CI: 1.87–1.92) in HICs and 1.97 (95% CI: 1.94–2.01) in LMICs.



- The overall PAFs for extreme maternal ages were higher in LMICs:
- Overall PAF for adolescents (≤ 19 years) was 2.4% (95% CI 2.3–2.5) in HICs versus 2.8 % (95% CI 2.7–3.1) in LMICs.
- Overall PAF for mothers ≥ 40 years was 2.9% (95% CI 2.9–3.0) in HICs versus 3.2% (95% CI 3.2–3.4) in LMICs.

Region	Total births	<15 years			15-19 years			Overall ≤ 19 years		
	(N)	Births (%)	RR	PAF (%)	Births (%)	RR	PAF (%)	Births (%)	RR	PAF (%)
HICs	143,845,444	0.1	2.32	0.1	6.1	1.39	2.3	6.2	1.4	2.4
LMICs	38,071,936	0.6	1.61	0.4	14.0	1.18	2.5	14.6	1.2	2.8

Region	Total births	20-24 years			25-29 years			30-34 years			35-39 years		
	(N)	Births (%)	RR	PAF (%)	Reference group			Births (%)	RR	PAF (%)	Births (%)	RR	PAF (%)
HICs	143,845,444	19.0	1.14	2.6				28.6	1.0	-0.3	14.7	1.27	3.8
LMICs	38,071,936	24.7	1.05	1.3				20.7	1.0	0.7	12.3	1.29	3.5

Region	Total births	40-44 years			≥ 45 years			Overall ≥ 40 years		
	(N)	Births (%)	RR	PAF (%)	Births (%)	RR	PAF (%)	Births (%)	RR	PAF (%)
HICs	143,845,444	3.2	1.86	2.6	0.2	2.32	0.3	3.4	1.89	2.9
LMICs	38,071,936	3.2	1.9	2.8	0.2	3.12	0.5	3.4	1.97	3.2

Conclusion

Extremes of maternal age are associated with increased stillbirth risk, with higher population impact in low- and middle-income settings.

These findings highlight the need for targeted strategies and action on structural inequities in maternal and newborn health.

References

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[Link to supplementary tables](#)